ORTAM developer exercise

Asbestos report application

Goal	An employee enters site information, reviews AI findings and downloads a report.
Core	A working prototype over two working days, up to 16 focused hours.
Bonus	PDF floor plan mapping, with separate optional time.

Fictional exercise. Detailed candidate brief defines the requirements.

Asbestos and sample records

Asbestos	Mineral fibres historically used in some building materials. This app records lab evidence.
Sample ID	A number such as S01 connects an employee location to a lab result.
Material zone	One measured material area. Several samples can describe the same zone.
Layer	One component, such as floor tile or the adhesive beneath it.
Quantity	The employee supplies the zone area or length. The lab does not measure it.

Laboratory result meanings

Detected	Asbestos appears in the tested layer. Keep its type and percentage.
Not detected	This result applies to the sampled layer. It does not describe the whole building.
Inconclusive	The lab could not give a definite result. Keep it unresolved.
Unsampled	No result exists. An inaccessible area stays unknown.

Employee inputs and AI inputs

Employee form	Building, rooms, material descriptions, sample IDs, quantities, condition and planned work.
Lab PDFs	AI extracts sample IDs, layers, result labels, reported types and percentages.
Exercise rules	The app uses the supplied review categories and fictional prices.
Word model	The generated report follows the supplied editable master structure.

The required employee journey

1 Enter and save	Create the project and material register. Reopen it after a reload.
2 Upload and review	Analyze the three lab PDFs. Show evidence and allow corrections.
3 Generate and edit	Match results, calculate costs and draft the report for review.
4 Preview and download	Display the report PDF and download matching Word and PDF files.

AI extraction with evidence

Structured fields	Sample ID, layer, result, type, percentage and source page.
Visible source	Show the PDF file, page and supporting text beside the extracted value.
Employee corrections	Let the employee fix a value before accepting it.
Visible uncertainty	Flag unmatched IDs, conflicts and extraction failures for review.

Cases that test the matching logic

Case	Expected behavior
S02 floor	Tile not detected, adhesive detected. Keep both layers.
S03 and S07	One ceiling zone. Charge its 140 square metres once.
S05 panel	Detected, quantity unknown. Cost remains Not priced.
S06 and M07	Inconclusive or unsampled. Keep them unresolved.

Recommendations use supplied rules

Category A	Detected with damage, poor condition or planned work. Flag priority professional review.
Category B	Other detected material. Flag professional management review.
Category C	Missing or inconclusive evidence. Explain the gap and follow up.
Category D	Not detected in the sampled material. Record its limited scope.

A transparent fictional estimate

Each measured zone	Quantity times the removal rate, plus quantity times the replacement rate.
Project allowances	Add CAD 2500 mobilization and CAD 1200 professional assessment once.
Contingency	Add 20 percent of the direct subtotal. Show the total before tax.
Missing quantity	Show Not priced and list the exclusion. Never substitute zero.

One report revision in both formats

Editable Word	Follow the supplied master and keep content editable.
Matching PDF	Convert that Word revision to PDF.
On-screen preview	Display that exact PDF so employees see the download layout.
After an edit	Mark the preview outdated, regenerate, then update both downloads.

Optional PDF floor plan mapping

Employee placement	Upload the plan, place a marker and assign a sample ID.
Linked result	Display its accepted laboratory status with text as well as color.
Stable position	Keep markers aligned when zooming, resizing and reopening.
Separate scope	CAD conversion and automatic geometry interpretation stay outside this test.

Two working days for the core

Day 1	Forms, persistence, PDF upload, real AI extraction and employee review.
Day 2	Matching, recommendations, costs, Word and PDF export, preview and verification.
Optional third day	Up to four additional hours for mapping if agreed in advance.
Time limit	Core maximum 16 focused hours. Record time and disclose unfinished work.

Submission and scoring

Core area	Points
Employee workflow	25
AI laboratory analysis	30
Report and recommendations	20
Cost calculations	15
Engineering and handover	10

Mapping bonus: up to 10 additional points, scored separately

Final candidate checklist

Working workflow	Enter the supplied site information, save it and analyze all three PDFs.
Correct findings	Keep layers, repeated samples, unknowns and evidence visible.
Usable output	Review the draft and open matching Word and PDF downloads.
Clear handover	Show the preview, explain setup, record time and disclose limitations.